

B.E. EVEN Mid-Semester Examinations 2024-25

Subject: Environmental Science | **Code:** MC-HU 402

1. Define sustainable development. How EIA can help in achieve sustainable development? (4)

Answer: Sustainable Development: Development meeting present needs without compromising the ability of future generations to meet theirs. Balances economy, environment, and social equity. **Role of EIA:** Environmental Impact Assessment predicts project consequences before they start. It helps achieve sustainability by identifying ecological damage early, suggesting cleaner tech, and recommending mitigation (like waste treatment) to align growth with stability.

2. a) What are the basic types of ecosystem? (2)

Answer: 1. Terrestrial Ecosystems: Land-based (Forests, Grasslands, Deserts). 2. Aquatic Ecosystems: Water-based (Lakes, Oceans, Coral Reefs).

2. b) Distinguish between abiotic and biotic components of an ecosystem. (3)

Answer: * **Abiotic:** Non-living physical/chemical elements (Sunlight, water, soil, air). They establish the environment's survivability. * **Biotic:** Living organisms (Plants, animals, bacteria). They interact for energy flow and nutrient cycling.

3. Describe energy flow model with the help of the laws of thermodynamics. (5)

Answer: Energy flow is unidirectional governed by: 1. **First Law:** Energy is conserved. Solar energy is captured by plants and transformed into chemical energy, then passed down the food chain. Total energy in universe remains constant. 2. **Second Law:** Energy transfer is inefficient. (10% Rule) As energy flows between trophic levels, ~90% is lost as metabolic heat. Only 10% transfers to the next level, limiting food chains to 4-5 levels.

4. a) Define green belt. (2)

Answer: A designated area of open land, forests, or parks preserved around urban areas where development is restricted to prevent sprawl, improve air quality, and reduce noise.

4. b) Describe the controlling measures of noise pollution. (3)

Answer: 1. Source: Silencers in vehicles/generators. 2. Transmission: Planting green belts, sound barriers on highways. 3. Receiver: Earplugs/PPE for industrial workers. 4. Legislative: Zoning laws keeping industries away from residential areas.

5. Explain how particulate matter (PM) contributes in air pollution. (5)

Answer: PM refers to microscopic suspended particles (PM₁₀ and PM_{2.5}) from exhausts, dust, and burning. * **Health:** PM_{2.5} penetrates deep into lungs and bloodstreams, causing asthma, bronchitis, and cardiovascular diseases. * **Visibility:** Scatters sunlight, creating smog and haziness. * **Environment:** Settles on soil/water altering pH, and on leaves hindering photosynthesis.